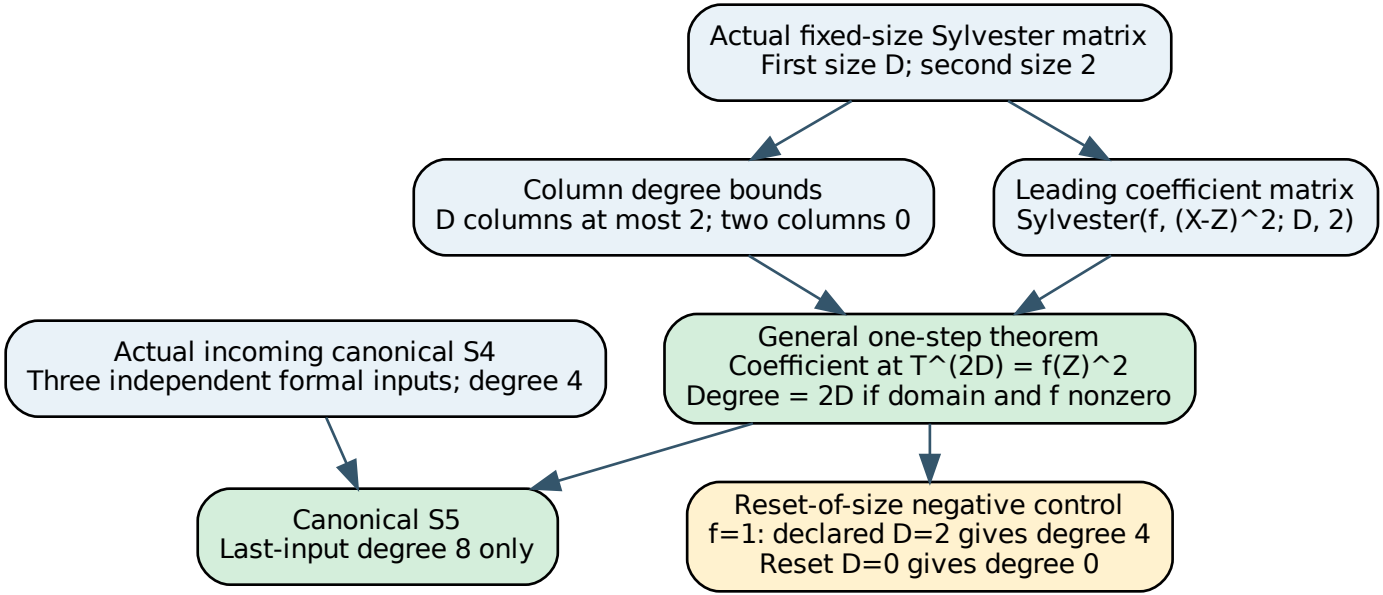
Actual fixed-size Sylvester matrix
First size D; second size 2
Column degree bounds
D columns at most 2; two columns 0
Leading coefficient matrix
Sylvester(f, (X-Z)^2; D, 2)
Actual incoming canonical S4
Three independent formal inputs; degree 4
General one-step theorem
Coefficient at T^(2D) = f(Z)^2
Degree = 2D if domain and f nonzero
Canonical S5
Last-input degree 8 only
Reset-of-size negative control
f=1: declared D=2 gives degree 4
Reset D=0 gives degree 0